GETTING STARTED

Getting Started with the ProtoCentral ADS1293 3-Channel ECG Breakout Board

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ProtoCentral ADS1293 3-channel, 24-bit ECG Breakout Board (with cables and electrodes)

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Introduction

The ProtoCentral ADS1293 breakout board is a 3-channel, 24-bit electrocardiogram (ECG) analog front-end based on the Texas Instruments ADS1293. It captures high-resolution cardiac signals from up to 3 simultaneous ECG channels, making it suitable for standard 3-lead and 5-lead ECG configurations. The board is designed for portable health monitoring, sports and fitness tracking, and research-grade biopotential measurement.

Each channel includes a dedicated 24-bit sigma-delta ADC with programmable sampling rates from 25.6 kHz down to 10.7 Hz, a built-in Right Leg Drive (RLD) amplifier for noise rejection, and a Wilson Central Terminal for precordial (chest)

lead measurements. The board connects to Arduino or ESP32 via the SPI bus, includes an onboard voltage regulator, and comes with ECG cables and disposable electrodes — everything you need to capture your first ECG waveform.

> **NOTE:**
> This board is intended for research and development purposes only. It is not FDA, CE, or FCC approved for consumer or medical use.

Key Features

- **3 simultaneous ECG channels** — 24-bit sigma-delta ADCs with independent channel programming
- **3-lead and 5-lead ECG support** — Einthoven limb leads plus optional precordial (V1) lead
- **Adjustable sampling rates** — 32, 64, 128, 256, ~533, or ~1066 samples per second
- **Built-in Right Leg Drive (RLD)** — active noise cancellation for cleaner signals
- **Wilson Central Terminal** — enables precordial lead measurements for 5-lead ECG
- **Ultra-low power** — 0.3 mW per channel, ideal for battery-powered applications
- **SPI interface** — high-speed digital communication with Arduino and ESP32
- **On-board 40 MHz clock** — no external crystal required
- **Onboard voltage regulator** — connect directly to 5V or 3.3V systems
- **Includes cables and electrodes** — 3-lead ECG cable with snap connectors and 10 disposable electrodes

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What's in the Box

- 1× ProtoCentral ADS1293 3-channel ECG breakout board
- 1× 3-lead ECG cable with snap connectors
- 10× Disposable ECG electrodes

You will also need an Arduino Uno (or compatible board), jumper wires, and a USB cable.

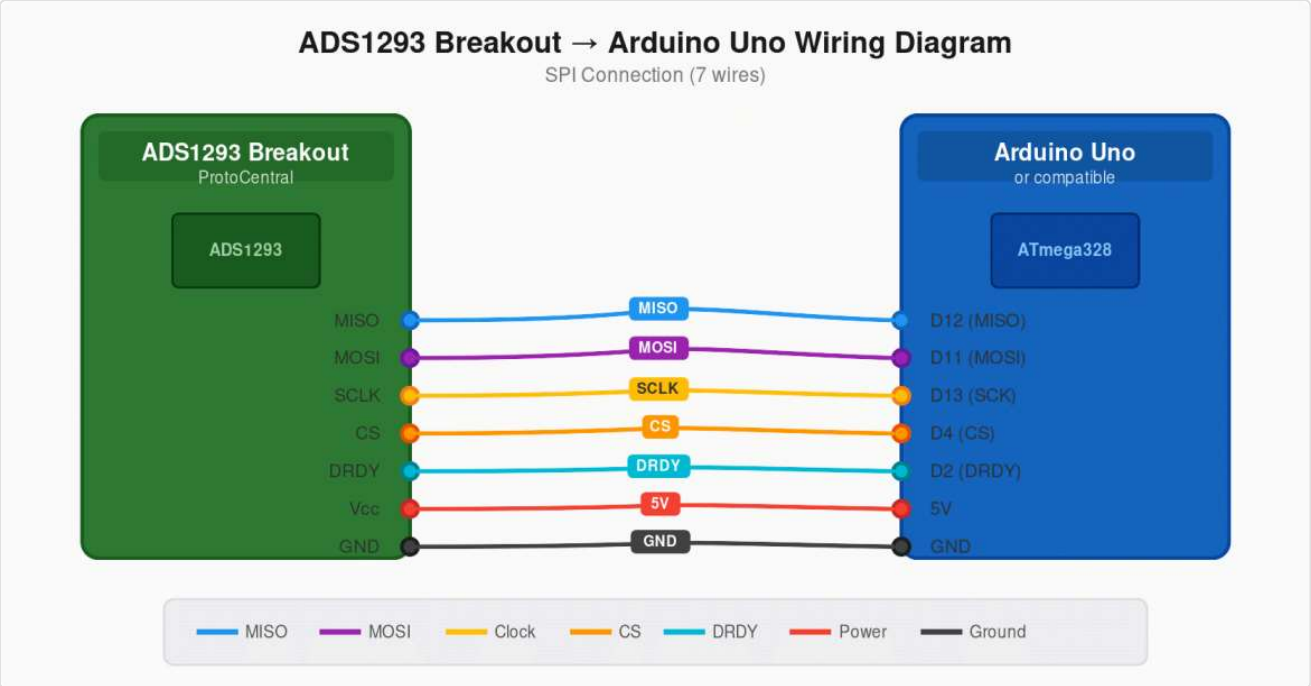
Specifications

PARAMETER	VALUE
ECG AFE IC	Texas Instruments ADS1293

PARAMETER	VALUE
ADC Resolution	24-bit per channel
Number of Channels	3 simultaneous + 1 pace channel
Sampling Rates	32, 64, 128, 256, ~533, ~1066 SPS
Differential Input Range	±400 mV
Interface	SPI
Power per Channel	0.3 mW
On-board Clock	40 MHz
Logic Level	3.3V or 5V (selectable via jumper)
Supply Voltage	3.3V to 5V (onboard regulator)
ECG Inputs	IN1–IN6 (6 electrode inputs)

Pin Connections

Wiring Diagram



ADS1293 Breakout to Arduino Uno

ADS1293 PIN	ARDUINO UNO PIN	FUNCTION
MISO	D12	SPI Data Out (Slave → Master)

ADS1293 PIN	ARDUINO UNO PIN	FUNCTION
MOSI	D11	SPI Data In (Master → Slave)
SCLK	D13	SPI Clock
CS	D4	Chip Select
DRDY	D2	Data Ready (interrupt)
Vcc	5V	Power
GND	GND	Ground

ADS1293 Breakout to ESP32

ADS1293 PIN	ESP32 PIN (VSPI)	FUNCTION
MISO	GPIO 19	SPI Data Out
MOSI	GPIO 23	SPI Data In
SCLK	GPIO 18	SPI Clock
CS	GPIO 4	Chip Select
DRDY	GPIO 2	Data Ready (interrupt)
Vcc	5V	Power
GND	GND	Ground

Electrode Connections

The ADS1293 breakout board has **two 3.5mm stereo (TRS) connectors** labeled **CON1** and **CON2**. Each connector carries 3 electrode wires (Tip, Ring, Sleeve), giving a total of 6 wires across the two connectors. Only 5 of these wires are used for electrodes — one wire on the CON2 connector is left unconnected.

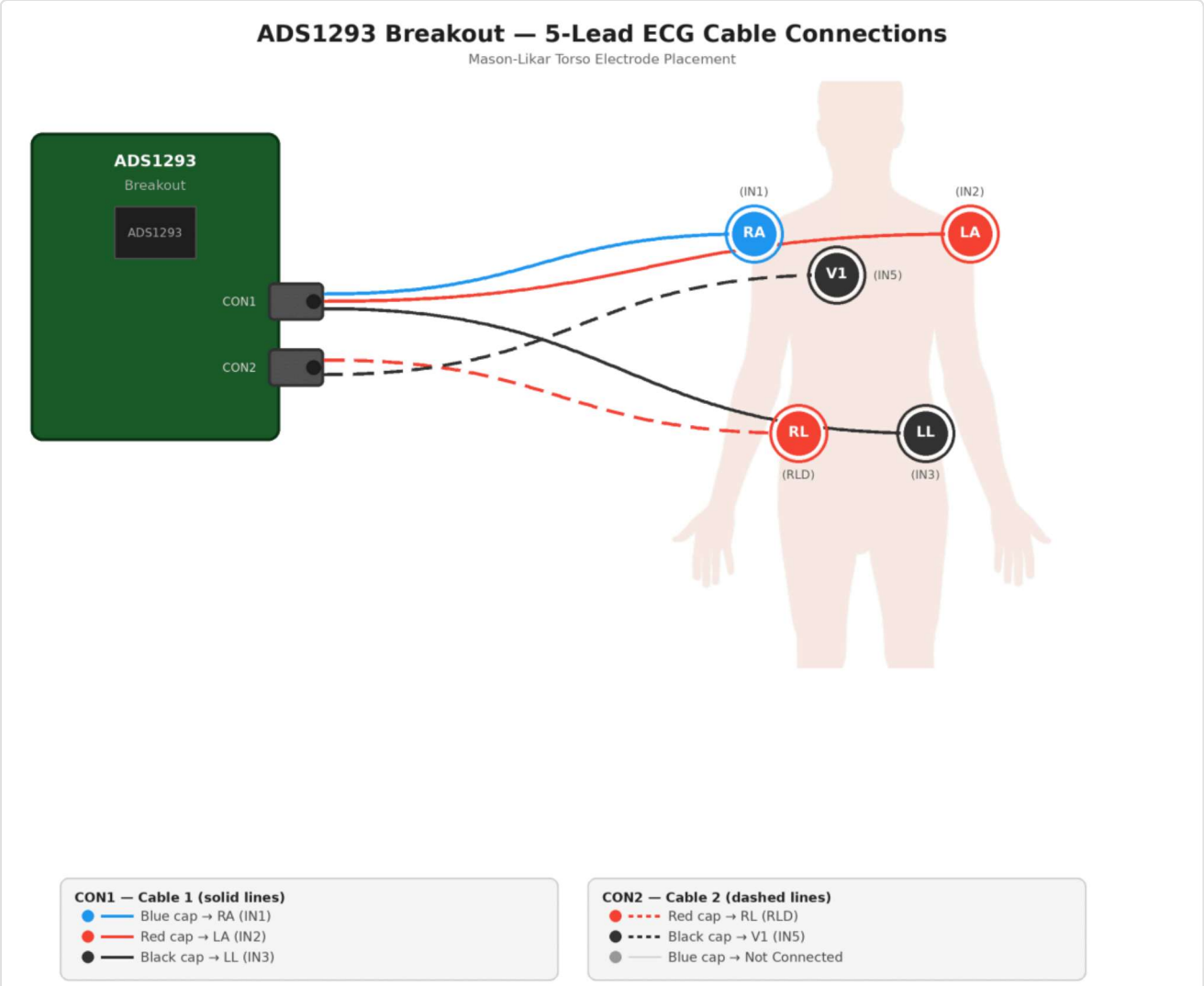
The board comes with **two 3-electrode snap cables**, each with a standard 3.5mm stereo plug on one end and three color-coded snap electrode leads on the other (Blue, Red, and Black caps).

> **IMPORTANT:**

> The cable lead labels printed on the snap connectors (LA, RA, RL) do **not** match the actual electrode placement. Always identify cables by the **color of the sticker on the cable cap** (Blue, Red, or Black) and follow the wiring tables below.

Electrode Placement Diagram

The diagram below shows the complete 5-lead ECG wiring with Mason-Likar torso electrode placement. Solid lines are Cable 1 (CON1) and dashed lines are Cable 2 (CON2).



3-Lead ECG (Einthoven Limb Leads)

For a basic 3-lead ECG, connect **one cable to CON1** only:

CABLE CAP COLOR	ELECTRODE PLACEMENT	ADS1293 INPUT	ECG LABEL
Blue	Below right collarbone	IN1	RA
Red	Below left collarbone	IN2	LA

CABLE CAP COLOR	ELECTRODE PLACEMENT	ADS1293 INPUT	ECG LABEL
Black	Lower left abdomen	IN3	LL

This configuration captures three standard limb leads:

- **Channel 1 → Lead I:** LA (IN2) minus RA (IN1)
- **Channel 2 → Lead II:** LL (IN3) minus RA (IN1)
- **Lead III** is derived mathematically: Lead II minus Lead I

> **NOTE:**
> In 3-lead mode, only CON1 is used. The Right Leg Drive (RLD) noise rejection is connected internally on the ADS1293. For best signal quality, add the RLD electrode using CON2 (see 5-lead setup below).

5-Lead ECG (with RLD and Precordial Lead)

For a 5-lead ECG, connect **both** cables:

Cable 1 → CON1 connector (solid lines in diagram):

CABLE CAP COLOR	ELECTRODE PLACEMENT	ADS1293 INPUT	ECG LABEL
Blue	Below right collarbone	IN1	RA
Red	Below left collarbone	IN2	LA
Black	Lower left abdomen	IN3	LL

Cable 2 → CON2 connector (dashed lines in diagram):

CABLE CAP COLOR	ELECTRODE PLACEMENT	ADS1293 INPUT	ECG LABEL
Red	Lower right abdomen	RLD	RL (Right Leg Drive)
Black	4th intercostal space, right sternal border	IN5	V1 (precordial)
Blue	Not connected	—	—

This adds two additional electrodes:

- **RL (RLD):** The Right Leg Drive electrode provides active noise cancellation, significantly improving signal quality on all channels.
- **Channel 3 → V1:** The precordial electrode at the 4th intercostal space (right sternal border) is measured against the Wilson Central Terminal reference, providing a chest lead view of the

heart.

Electrode Placement Guide (Mason-Likar)

All electrodes are placed on the **torso** (not on the limbs), following the Mason-Likar placement used in clinical settings:

ECG LABEL	POSITION ON BODY
RA	Below the right collarbone, near the right shoulder
LA	Below the left collarbone, near the left shoulder
RL	Lower right abdomen
LL	Lower left abdomen
V1	4th intercostal space, right sternal border (between ribs, next to the sternum)

> **TIP:**
> Use the disposable snap electrodes on clean, dry skin. Wipe the electrode sites with an alcohol swab for best adhesion and signal quality.

Installing the Arduino Library

Option 1: Arduino Library Manager (Recommended)

1. Open the Arduino IDE
2. Go to **Sketch → Include Library → Manage Libraries...**
3. Search for "**Protocentral ADS1293**"
4. Find "**ProtoCentral ADS1293 ECG library**" and click **Install**

Option 2: Manual Install from GitHub

1. Go to github.com/Protocentral/protocentral-ads1293-arduino
2. Click **Code → Download ZIP**
3. In the Arduino IDE, go to **Sketch → Include Library → Add .ZIP Library...**
4. Select the downloaded ZIP file

Your First ECG Reading (3-Lead)

Open the example sketch: **File → Examples → ProtoCentral ADS1293 → 3-lead ECG streaming**

Or create a new sketch with the following code:

```
#include "protocentral_ads1293.h"
#include

#define DRDY_PIN 2
#define CS_PIN 4

ADS1293 ecg(DRDY_PIN, CS_PIN);

void setup() {
    Serial.begin(115200);
    ecg.begin();

    ecg.configureChannel1(FlexCh1Mode::Default);
    ecg.configureChannel2(FlexCh2Mode::Default);
    ecg.enableCommonModeDetection(CMDetMode::Enabled);
    ecg.configureRLD(RLDMode::Default);
    ecg.setSamplingRate(ADS1293::SamplingRate::SPS_128);
    ecg.configureChannelConfig(ChannelConfig::Default3Lead);
    ecg.applyGlobalConfig(GlobalConfig::Start);
}

void loop() {
    if (ecg.isDataReady()) {
        auto samples = ecg.getECGData();
        if (samples.ok) {
            Serial.print(samples.ch1);
            Serial.print(',');
            Serial.print(samples.ch2);
            Serial.print(',');
            Serial.println(samples.ch3);
        }
    }
}
```

What This Code Does

1. **Includes the libraries** — `protocentral_ads1293.h` for the ECG front-end and `SPI.h` for SPI bus communication
2. **Creates the ECG object** — with DRDY on pin 2 and CS on pin 4
3. **Configures channels** — sets up channels 1 and 2 for default 3-lead ECG, enables common-mode detection and Right Leg Drive for noise rejection
4. **Sets sampling rate** — 128 samples per second (good balance of resolution and data rate)
5. **Starts acquisition** — `applyGlobalConfig(GlobalConfig::Start)` begins continuous ECG capture
6. **Reads data** — checks `isDataReady()` each loop iteration, then prints comma-separated channel values for the Arduino Serial Plotter

Using the Arduino Serial Plotter

1. Upload the sketch to your Arduino
2. Attach the 3 ECG electrodes to your torso: RA (below right collarbone), LA (below left collarbone), LL (lower left abdomen)
3. Connect the electrode cable snaps to the electrodes
4. Open **Tools → Serial Plotter** at **115200 baud**
5. You should see live ECG waveforms — the characteristic PQRST complex of your heartbeat

> **TIP:**
> For cleaner signals, place electrodes on smooth skin with minimal hair. Clean the skin with an alcohol wipe before attaching the electrode.

ESP32 Setup

For ESP32, specify custom SPI pins in `begin()`:

```
#define SCK_PIN 18
#define MISO_PIN 19
#define MOSI_PIN 23

ecg.begin(SCK_PIN, MISO_PIN, MOSI_PIN);
```

Visualizing with OpenView

For a richer visualization experience, use the ProtoCentral OpenView application:

1. Install OpenView 2 — see the setup guide for downloads + first run on Windows / Linux / iOS / Android
2. Upload the **OpenView example** sketch from the library examples
3. Open OpenView 2, select "**ADS1293 breakout**" from the board dropdown
4. Select the correct serial port
5. Click **Connect** to see real-time ECG waveforms with heart rate

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Troubleshooting

No ECG signal / flat line

- Check that electrodes are properly attached to skin with good contact

- Verify SPI wiring — MISO, MOSI, SCLK, CS, and DRDY must all be correctly connected
- Ensure the electrode cable snaps are firmly connected to the breakout board's electrode port
- Check that DRDY is on an interrupt-capable pin (D2 on Arduino Uno)

Very noisy signal

- Make sure the Right Leg Drive electrode (RL) is connected — this is essential for noise rejection
- Keep electrode wires away from power supplies and motors
- Ensure the subject is not touching grounded metal objects
- Try a lower sampling rate (64 or 32 SPS) for better noise filtering
- Replace electrodes if they have been attached for a long time (gel dries out)

Only one or two channels working

- Check that all electrode connections are secure
- For 3-lead ECG, you need at minimum: RA, LA, LL, and RLD connected
- Verify that all 3 channels are configured in your sketch (Channel 1, 2, and optionally 3)

SPI communication errors

- Verify CS and DRDY pins in your code match your physical wiring
- Check that no other SPI devices are conflicting on the bus
- Try reducing SPI clock speed if using long jumper wires

Resources

- **Arduino Library:** github.com/Protocentral/protocentral-ads1293-arduino
- **Hardware Design Files:** GitHub repository
- **ADS1293 Datasheet:** TI ADS1293 Datasheet (PDF)
- **OpenView 2:** setup guide · source on GitHub

Licenses

- **Hardware:** Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0)
- **Software:** MIT License

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